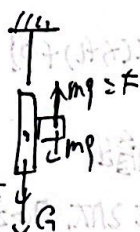


油 (6 普物)



$$G + F = (m + m)g = mg$$

$$\therefore a = \frac{(m + m)g}{m}$$

2.

$$\frac{1}{2} K L^2 = \frac{1}{2} m v^2$$

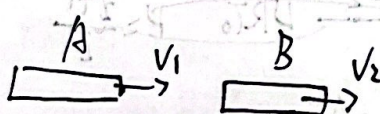
$$\textcircled{1} v = \sqrt{\frac{2 K L^2}{m}}$$

$$M v = (n m + m) v'$$

$$v' = \frac{m}{n m + m} \cdot \sqrt{\frac{2 K L^2}{m}}$$

$$= \frac{\sqrt{2 K L^2}}{n + 1}$$

3. 4. 5. 6.



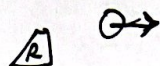
$$L' = (v_2 - v_1) \Delta t$$

$$L' = L \sqrt{1 - \left(\frac{v_2}{c}\right)^2}$$

$$v_1 \Delta t = L \sqrt{1 - \left(\frac{v_2}{c}\right)^2}$$

$$v_2 = \frac{c^2 L^2}{L^2 + c^2 \Delta t^2}$$

7.



$$v_s = f \cdot \lambda_s = 340 \text{ m/s}$$

$$\text{由 } \lambda_k = \frac{v_a + v_r}{v_a - v_s} \lambda_s$$

$$\lambda_1 = \frac{v_a}{v_a + v_s} \lambda_s = \frac{340}{340 + v_s} \lambda$$

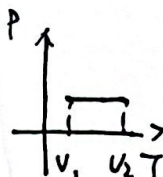
$$\lambda_2 = \frac{v_a}{v_a - v_s} \lambda_s = \frac{340}{340 - v_s} \lambda$$

$$\lambda_k = |\lambda_1 - \lambda_2|$$

$$3 = \left(\frac{340}{340 - v_s} - \frac{340}{340 + v_s} \right) \lambda$$

$$v_s = \sqrt{\frac{340^2 - 680 \lambda}{3}}$$

8.



$$\Delta S = \int_{T_1}^{T_2} \frac{dQ}{T}, \quad dQ = n C_{pm} dT$$

$$= \int_{T_1}^{T_2} \frac{C_{pm} dT}{T} = C_{pm} \ln \frac{T_2}{T_1}$$

$$= C_{pm} \ln \frac{T_2}{T_1}$$

$$= C_{pm} \ln 2$$

9. 10. 11.

$$\text{11. } \vec{I} \rightarrow \vec{E} \rightarrow \vec{B}$$

$$\vec{B} = \frac{\mu_0 I}{2R} \quad (\text{for } r < R)$$

$$\Phi = \vec{B} \cdot \vec{S}$$

$$= \frac{\mu_0 I}{2R} \cdot \pi r^2 \cdot \cos \omega t$$

$$\mathcal{E} = - \frac{d\Phi}{dt} = \frac{\omega \mu_0 I \pi r^2}{2R} \sin \omega t$$

$$I = \frac{\mathcal{E}}{R_0} = \frac{\omega \mu_0 I \pi r^2}{2R \cdot R_0} \sin \omega t$$



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二、计算题

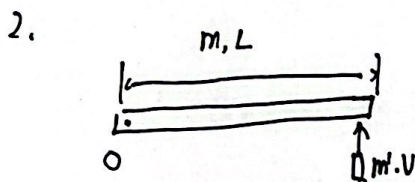


$$m_1 g R - M_f = J \alpha, \quad m_2 g R - M_f = J \alpha$$

$$h = \frac{1}{2} a_1 t_1^2 = \frac{1}{2} a_2 t_2^2, \quad J = \frac{(m_2 - m_1) g R^2}{(a_2 - a_1)}$$

$$a_1 = \frac{2h}{t_1^2}, \quad a_2 = \frac{2h}{t_2^2}$$

$$\therefore J = \frac{(m_2 - m_1) g R^2 t_1^2 t_2^2}{2h(t_2^2 - t_1^2)}$$



(1) 由角动量守恒:

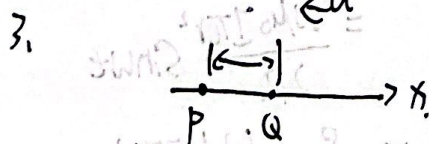
$$J' \omega' = (J' + J) \omega, \quad J = \frac{1}{3} m L^2, \quad J' = m L^2$$

$$\omega' = \frac{v}{L}$$

$$\therefore \omega = \frac{m' L v}{m' L^2 + \frac{1}{3} m L^2}$$

$$(2) \quad \frac{1}{2} (J' + J) \omega^2 = \int_0^\theta M_r d\theta$$

$$\theta = \frac{(J' + J) \omega^2}{2 M_r}$$



$$(1) \quad y_0 = A \cos(\omega t + \phi)$$

$$y_0 = A \cos[\omega(t + \frac{L}{u}) + \phi]$$

$$y_0 = A \cos(\omega t + kL + \phi)$$

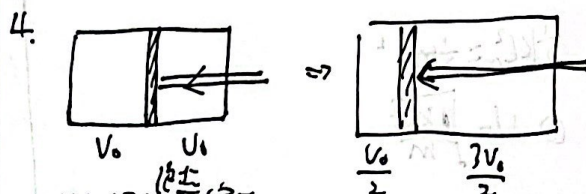
$$k = \frac{2\pi}{\lambda} = \frac{\omega}{u}$$

$$(2) \quad y_0 = A \cos(\omega t + kx + \phi + kL)$$

$$= A \cos[\omega t + k(x + L) + \phi]$$

(3) 由波动表达式可看出
当 $k(x + L) = 2n\pi$, $n = 0, 1, 2, \dots$ 为波节

$$x = \frac{2n\pi u}{\omega} - L$$



可逆过程的压强:

$$W_I = \frac{P_1 V_1 - P_2 V_2}{\mu - 1}$$

$$= \frac{1 - 2^{\mu-1}}{\mu - 1} \cdot \mu R T_0$$

$$\Delta E_I = -W_I, \quad \Delta E_{II} = -W_{II}$$

$$W_{II} = \frac{P_1 V_1 - P_2 V_2}{\mu - 1}$$

$$= \frac{1 - (\frac{2}{3})^{\mu-1}}{\mu - 1} \cdot \mu R T_0$$

$$W_F = W_I - W_{II}$$

$$= \frac{(\frac{2}{3})^{\mu-1} - 2^{\mu-1}}{\mu - 1} \cdot \mu R T_0$$

$$W_F = \Delta E_I + \Delta E_{II}$$

$$= \frac{2^{\mu-1} + (\frac{2}{3})^{\mu-1} - 2}{\mu - 1} \cdot \mu R T_0$$

5. 设向径为 r_0

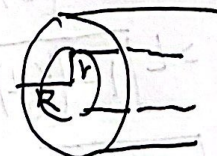
设单位长度带电为 Q

$$\oint E \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{2\pi \epsilon_0 r}$$

$$\oint E \cdot d\vec{s} = Q$$

$$h = \frac{Q}{2\pi r}$$



$$E_{\max} = \frac{Q}{2\pi \epsilon_0 r_0}$$

$$Q = 2\pi \epsilon_0 r_0 E$$

$$u = \int_{r_0}^R E \cdot d\vec{r} = r_0 E \ln \frac{R}{r_0}$$



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6.

$$N=5000$$

$$a=b$$

- ① 斜入射
② 垂直入射

(1)

$$d=a+b=\frac{\Delta L}{N}=2 \times 10^{-6} \text{ m}$$

$$d \sin \theta = k \lambda \quad (\text{明纹})$$

$$\text{当 } \theta = \frac{\pi}{2} \text{ 时}$$

$$k = \frac{d}{\lambda} = \frac{2 \times 10^{-6}}{5.89 \times 10^{-7} \text{ m}}$$

∴

$$\text{而由 } d \sin \theta = k \lambda, k_1 = 4, 2, \dots$$

$$d \sin \theta = k_2 \lambda, k_2 = 0, 1, 2$$

$$k_2 = 2k_1$$

故 $k_2 = 2, 4, 6, \dots$ 为缺级

∴ 共能看到 ~~2k+1-2=5~~ 条

即 ~~级次~~ 级次为 ~~0, ±1, ±3~~ 0, ±1, ±3

(2)

$$d(\sin \theta - \sin \theta_0) = k \lambda$$

$$d(\sin \theta - \sin \theta_0) = k \lambda$$

$$d(\sin \theta - \sin \theta_0) = k \lambda$$

$$\text{当 } \theta = \frac{\pi}{2} \text{ 时}$$

$$\text{解得 } k_1 = \frac{d}{2\lambda} \approx 1$$

$$k_2 = -\frac{3d}{2\lambda} \approx -6$$

同样 $k_2 = \pm 2, \pm 4, \pm 6, \dots$ 为缺级

故级数为 5 条, 级次依次为 -6, -5, -4, -3, -2, -1, 0, 1

故级数为 5 条, 级次依次为 -5, -3, -1, 0, 1

7.

① 在边区域, 单位面积上的力为:

$$F_1 = -\frac{\partial W_0}{\partial x}, \quad dW_0 = \frac{1}{2} \vec{B} \cdot \vec{H} \cdot d\vec{h}$$

$$\therefore F_1 = -\frac{B_1^2}{2\mu_0} \quad (\text{沿 } x \text{ 轴方向})$$

$$\text{同理 } F_2 = +\frac{B_2^2}{2\mu_0} \quad (\text{沿 } x \text{ 轴方向})$$

① 图中 $B_2 > B_1$

$$\therefore F = F_2 - F_1$$

$$= \frac{B_2^2 - B_1^2}{2\mu_0} \quad (\text{方向向右})$$



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